Physics A (H156, H556)

Exploring - Electrical potential
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **42**.
- The total number of marks may take into account some 'either/or' question choices.

1 This question is about lightning.

Fork lightning is an electrical discharge that occurs between the bottom of the cloud and the surface of the Earth.

A cloud has a charge of 155 C and is at a height of 2.0 km.

The surface of the Earth has an electrical potential V of 0 V.

i. Assume the cloud acts as a **point** charge.

Calculate the magnitude of the electrical potential V between the cloud and the surface of the Earth.

V = ... V **[2]**

ii. A fork lightning strike has a duration of 25 ms. The cloud discharges at a constant rate. The cloud is uncharged after the strike.

Calculate the number of electrons reaching the ground in 1.0 ms.

number of electrons in 1.0 ms = ... **[3]**

2 The diagram below shows two oppositely charged spheres.

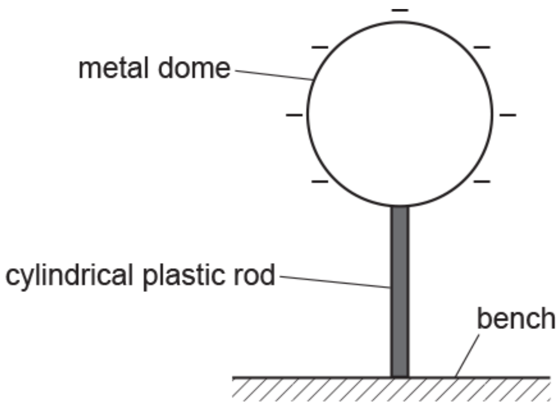
The magnitude of the charge on each sphere is the same.
The point **P** is on the line joining the centres of the spheres and is the same distance from the centre of each sphere.

Which statement is correct?

A A negatively charged particle at **P** will move to the right.
B The direction of the electric field at **P** is to the left.
C The electric potential at **P** is zero.
D The magnitude of the electric field strength at **P** is zero.

Your answer [ ] **[1]**

3 A spherical metal dome shown below is charged to a potential of –12 kV.



The dome is supported by a cylindrical plastic rod. The radius of the dome is 0.19 m.

i. Show that the magnitude of the total charge Q on the dome is 2.5×10^{-7} C.

[2]

ii. The dome discharges slowly through the plastic rod.
It takes 78 hours for the dome to completely discharge.

1 Show that the mean current I in the plastic rod is about 9×10^{-13} A.

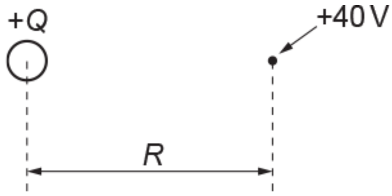
[2]

2 The average potential difference across the plastic rod during discharge is 6000 V.
The rod has cross-sectional area 1.1×10^{-4} m^2 and length 0.38 m.

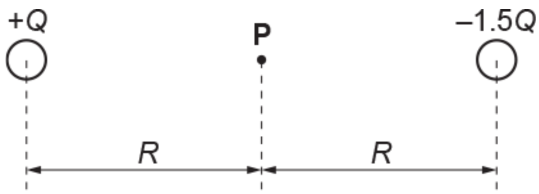
Calculate the resistivity ρ of the plastic.

ρ = ... Ωm **[3]**

4 The electric potential at a distance R from the centre of a charge $+Q$ is + 40 V.



What is the potential at the point **P** for the arrangement of the charges $+Q$ and $-1.5Q$ as shown below?



A – 20 V

B – 60 V

C + 80 V

D + 100 V

Your answer [1]

5

A proton with kinetic energy 0.52 MeV is travelling directly towards a stationary nucleus of cobalt-59 ($^{59}_{27}Co$) in a head-on collision.

i. Explain what happens to the electric potential energy of the proton-nucleus system.

____

____

____ **[1]**

ii. Calculate the **minimum** distance *R* between the proton and cobalt nucleus.

R = ____________________________ m **[3]**

6 An isolated metal sphere is charged using a power supply.

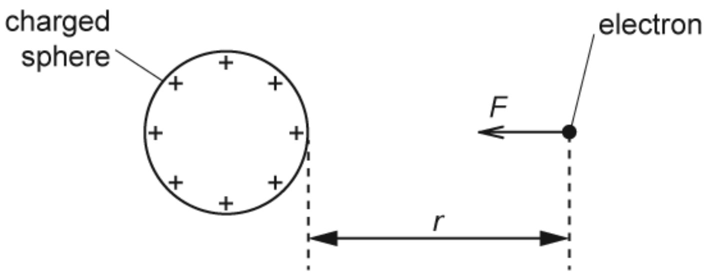
Which single quantity can be used to determine the capacitance of the sphere?

A The diameter of the sphere.
B The charge on the sphere.
C The resistance of the metal.
D The e.m.f. of the power supply.

Your answer

[1]

7 An electron is released at a distance r from the surface of a positively charged sphere. It is attracted towards the centre of the sphere and moves until it touches the surface.



Which of the following statements is/are correct?

1 The area under the F against r graph is equal to work done on the electron.

2 The electric field strength E at distance r is equal to $\frac{F}{1.6 \times 10^{-19}}$.

3 The work done on the electron is equal to $F \times r$.

A Only 1
B Only 1 and 2
C Only 1 and 3
D 1, 2 and 3

Your answer [1]

8(a)

Fig. 20.1 shows a positively charged metal sphere and a negatively charged metal plate.

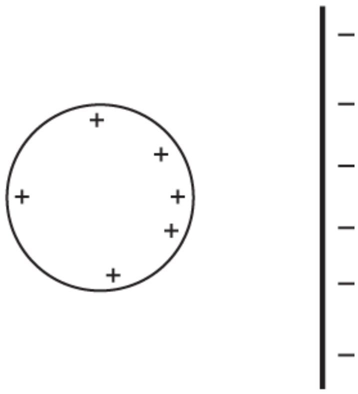


Fig. 20.1

On Fig. 20.1, draw a minimum of **five** electric field lines to show the field pattern between the sphere and the plate.

[2]

(b) Define *electric potential* at a point in space.

[1]

(c) A metal sphere is given a positive charge by connecting its surface briefly to the positive terminal of a power supply. The electric potential at the surface of the sphere is + 5.0 kV. The sphere has radius 1.5 cm.

i. Show that the charge Q on the surface of the sphere is 8.3×10^{-9} C.

[2]

ii. Fig. 20.2 shows the charged sphere from **(i)** suspended from a nylon thread and placed between two oppositely charged vertical plates.

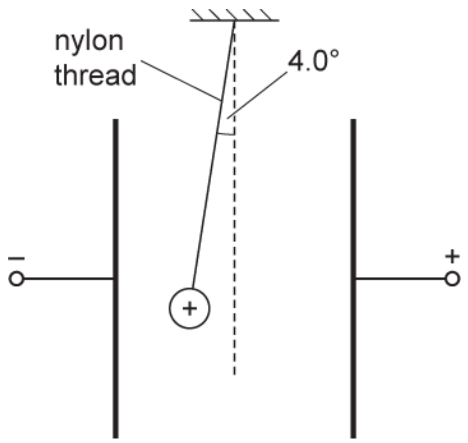


Fig. 20.2 (not to scale)

The weight of the sphere is 1.7×10^{-2} N. The string makes an angle of 4.0° with the vertical.

1. Show that the electric force on the charged sphere is 1.2×10^{-3} N.

[1]

2. Calculate the uniform electric field strength *E* between the parallel plates.

E = ___________________ N C^{-1} **[2]**

9 Fig. 21.1 shows two oppositely charged ions to the left of a point **X**.

Fig. 21.1

The separation between the centres of the ions is 3.0×10^{-10} m. Each ion has charge of magnitude 1.6×10^{-19} C.

i. Explain why the direction of the **resultant** electric field strength at point **X** is to the left.

____

____

____

____ **[2]**

ii. Calculate the minimum energy in eV required to completely separate the ions.

energy = ______________________________ eV **[3]**

10(a) Fig. 21.2 shows a uniformly charged sphere of radius R.

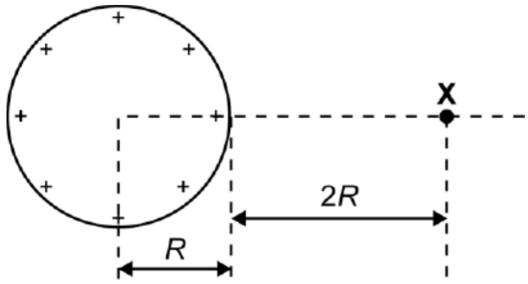


Fig. 21.2

The electric potential at point **X** is +1800 V. Point **X** is at a distance of $2R$ from the **surface** of the sphere.

i. Calculate the electric potential V at the surface of the sphere.

V = ____________________ V **[2]**

ii. The radius of the sphere is 4.0 cm.

Calculate

1. the surface charge Q on the sphere

Q = ____________________ C **[2]**

2. the electric field strength E at the surface of the sphere.

E = ____________________ N C^{-1} **[2]**

(b) Fig. 21.3 shows two particles with the same charge but of opposite sign.

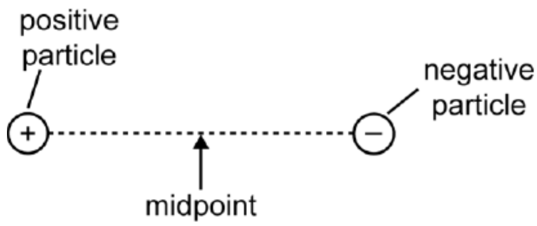


Fig. 21.3

State and explain the magnitude of the electric potential at the midpoint between the particles.

--- **[2]**

11 The electric potential is -1.2×10^{-4} J C^{-1} at a point 1.2×10^{-5} m from an isolated electron.

An α-particle $^{4}_{2}$He passes through this point.

What is the magnitude of the electric potential at the mid-point between the α-particle and the electron at this instant?

A. -7.2×10^{-4} J C^{-1}
B. $+2.4 \times 10^{-4}$ J C^{-1}
C. $+4.8 \times 10^{-4}$ J C^{-1}
D. $+7.2 \times 10^{-4}$ J C^{-1}

Your answer ☐

[1]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1		i	$V = \dfrac{Q}{4\pi\varepsilon_0 r}$ $V = \dfrac{155}{(4\pi\ 8.85\times10^{-12} \times 2\times10^{3})}$ = 7.0 x 10^8 (V)	C1 A1	All values substituted correctly Correct answer to at least 2sf (6.97) __Examiner's Comments__ Most candidates were able to correctly select the equation and evaluate the correct answer. Some chose the field strength equation or wrote the potential equation with a r^2. Some selected the wrong value of r, giving it as the radius of the cloud. As with Question 18(b)(iii) candidates are reminded that, in general, answers should be given to the lowest number of significant figures in the question, and that an answer of 7 × 10^8 will incur the significant figure penalty (applied only once per paper).
		ii	Either (I =) 155 / 25 × 10^{-3} = (6200 A) n = 6200 / 1.6 × 10^{-19} = 3.88 x10^{22} n (in 1ms) = 3.88 × 10^{22} × 10^{-3} = 3.9 × 10^{19} Or (Charge flow) = 155/25 = (6.2 C ms^{-1}) n = 6.2 / 1.6 × 10^{-19} n = 3.9 × 10^{19}	C1 C1 A1 (C1) (C1) (A1)	**Ignore** units throughout although penalise any unit on answer line Correct evaluation to at least 2sf (3.88) Correct evaluation to at least 2sf (3.88) __Examiner's Comments__ This question could be answered in several ways. Most candidates chose to calculate the current in amps and then converting back to ms at the end. Very few were confused by the need to give the answer in ms. This question polarised the candidates with the vast majority scoring 0 or 3; those who got part way to solving it generally ended up with the correct answer.
			Total	5	

Question			Answer/Indicative content	Marks	Guidance
2			C	1	<u>Examiner's Comments</u> The correct response is **C**. Candidates often find the electric field questions challenging and this was again the case as this question was correctly answered by only one third of the candidates. Many candidates drew arrows on the diagram to assist them. Response **D** was the most common distractor; linking this to gravitational fields would produce a zero field strength at **P** which is likely to be the reason.
			Total	1	
3		i	$12\,000 = \frac{Q}{4\pi\varepsilon_0 r}$ $12\,000 = \frac{Q}{4\pi\varepsilon_0 \times 0.19}$ $Q = 2.5(4) \times 10^{-7}$ (C)	C1 C1 A0	**Allow** $E = (V / d =)\ 6.316 \times 10^4$ and **C1** $E = 6.316 \times 10^4 = \frac{Q}{4\pi\varepsilon_0 \times 0.19^2}$ **C1**
		ii	**1** $t = 78 \times 3600$ $(I =) \frac{2.5 \times 10^{-7}}{78 \times 3600}$ $I = 8.9 \times 10^{-13}$ (A) **2** $(R =) \frac{6000}{9.0 \times 10^{-13}}$ **or** 6.7×10^{15} (Ω) **or** $V = IR$ and $R = \frac{\rho L}{A}$ $\frac{6000}{9.0 \times 10^{-13}} = \frac{\rho \times 0.38}{1.1 \times 10^{-4}}$ $\rho = 1.9 \times 10^{12}$ (Ω m)	C1 C1 A0 C1 C1 A1	There is no ECF from **(b)(i)** **Note** 2.54×10^{-7} gives an answer 9.0×10^{-13} A There is no ECF from **(b)(ii)1** **Take** 12000 V as **TE** for this C1 mark, then ECF **Note** 8.9×10^{-13} (A) gives an answer 2.0×10^{12} (Ω m)
			Total	7	
4			A	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
5		i	Kinetic energy (of proton) changes to potential (energy) **or** Potential energy increases as the kinetic energy (of the proton) decreases **or** Potential energy increases as work is done against the field / against repulsion / positive charge	**B1**	**Allow** 'it' / PE for (electric) potential energy **Allow** KE / E_k

Question			Answer/Indicative content	Marks	Guidance
		ii	energy = $0.52 \times 10^{6} \times 1.60 \times 10^{-19}$ **or** $8.3(2) \times 10^{-14}$ (J)	C1	
			$\frac{1.60 \times 10^{-19} \times 27 \times 1.60 \times 10^{-19}}{4\pi\varepsilon_0 R} = 8.32 \times 10^{-14}$	C1	
			$R = 7.5 \times 10^{-14}$ (m)	A1	**Allow** 2 mark for 1.6×10^{-13} (m); Z = 59 used **Allow** 2 mark for 8.9×10^{-14} (m); Z = 32 used **Allow** 1 mark for 2.8×10^{-15} (m); Z = 1 used **Allow** 1 mark for 1.2×10^{-32} (m); energy = 5.2×10^{5} used **Examiner's Comments** The above question on electric potential energy provided excellent discrimination with middle and upper quartile candidates showing how to produce immaculate answers – identify the physics, write down the correct physical equation, do any necessary conversions (e.g. MeV to J), rearrange the equation, substitute correctly and then write the final answer in standard form to the correct number of significant figures. About a third of the candidates scored full marks. Some of the missed opportunities or errors were: • Using an incorrect equation with the distance squared • Not correctly converting the kinetic energy 0.52 MeV into joule (J) • Using the equation $r = r_0A^{1/3}$ for the mean radius of a nucleus to determine the minimum distance
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
6			A	1	
			Total	1	
7			B	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
8	a		Correct pattern Correct direction of the field	B1 B1	**Note:** At least five field lines must be drawn and of these, two must be perpendicular (by eye) to the surface of the sphere and plate **Note:** This may be shown on just one line **Examiner's Comment** Most candidates drew decent field patterns and showed the correct direction of the electric field. It is difficult to draw curved field lines, but those who were careful and had the field lines perpendicular at both the surface of the sphere and the metal plate were rewarded.
	b		(Electric potential) is the <u>work</u> done per (unit) charge in bringing a <u>positive</u> charge from infinity (to the point).	B1	**Allow:** <u>work</u> done / <u>energy</u> required to bring a unit <u>positive</u> charge from infinity (to the point) **Examiner's Comment** This was not well-answered; the modal mark was zero. Definition for electric potential lacked precision and often made no reference to a 'unit **positive** charge' or 'per unit **positive** charge'. At times, other quantities such as electric field strength and gravitational field strength were being defined. This was a missed opportunity -definitions just need to be learnt.
	c	i	$V = Q/4\pi\varepsilon_0 r$ (Allow any subject) $Q = 4\pi \times 8.85 \times 10^{-12} \times 0.015 \times 5000$ $Q = 8.3(4) \times 10^{-9}$ (C)	C1 C1 A0	**Note** using $E = V/d$ with $E = \Omega/4\pi\varepsilon_0 r^2$ is wrong physics and hence scores zero **Note** if the value of ε_0 is not given here, it could be implied in the correct 3sf answer **Allow** any subject here if the answer is given to more than 2sf **Allow** the use of $1/4\pi\varepsilon_0 = 9 \times 10^9$ **Examiner's Comment** By contrast to the last question, the answers here were perfect. Correct values were substituted into the equation for electric potential to show that the charge was that stated in the question. In a 'show' question, always give the final answer to more significant figures than the required answer. It was good to see many scripts with the final answer written as 8.3**4** × 10^{-9} C.

Question			Answer/Indicative content	Marks	Guidance
		ii	1 (electric force =) $1.7 \times 10^{-2} \times \tan 4.0$ (Allow any subject) (electric force = 1.19×10^{-3} N) 2 $E = 1.2 \times 10^{-3}/8.3(4) \times 10^{-9}$ $E = 1.4 \times 10^{5}$ (N C^{-1})	M1 (A0) C1 A1	**Not** 1.7×10^{-2} sin4 or 1.7×10^{-2} cos86 **Allow** $1.7 \times 10^{-2} \times$ sin4/cos4 **Allow** 2 marks for 1.45×10^{5} (N C^{-1}), 8.3×10^{-9} used **Allow** 2 marks for 1.43×10^{5} (N C^{-1}), 1.19×10^{-3} (N) used **Examiner's Comment** This was a good discriminator with high-scoring candidates either using triangle of forces, or resolution of forces, to determine the electric force on the sphere. The value of the force was given so that it could be used to answer the next question. More than half of the candidates correctly calculated the electric field strength using the information provided in **(c)(i)** and **(c)(ii)1.** Some candidates used the elementary charge rather than the value from **(c)(i)** to calculate the field strength; this gave an incorrect answer of 7.5×10^{15} N C^{-1}.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
9		i	The direction of the electric field due to the negative charge is to the left and right for the positive charge.	B1	
		i	The magnitude of the electric field strength due to the positive charge is smaller than that for the negative charge (because of greater distance). (Hence the resultant electric field strength is to the left.)	B1	
		ii	energy = $\dfrac{Qq}{4\pi\varepsilon_0 r} = \dfrac{(1.60\times10^{-19})^2}{4\pi\varepsilon_0 \times 3.0\times10^{-10}}$	C1	
		ii	energy = $7.67(2) \times 10^{-19}$ (J)	C1	
		ii	energy = 4.8 (eV)	A1	
			Total	**5**	
10	a	i	$V \propto 1/r$ or distance = $3R$	C1	
		i	V = 5400 (V)	A1	
		ii	1 $5400 = \dfrac{Q}{4\pi\times8.85\times10^{-12}\times0.04}$ (Any subject)	C1	Possible ecf from **(i)**
		ii	$Q = 2.4 \times 10^{-8}$ (C)	A1	
		ii	2 $E = \dfrac{2.4\times10^{-8}}{4\pi\times8.85\times10^{-12}\times0.04^2}$	C1	Possible ecf from **(ii)1**
		ii	$E = 1.35 \times 10^5$ (N C^{-1})	A1	
	b		The magnitude of the electric potential is the same for both particles at the midpoint but of opposite sign.	B1	
			The (total) potential at the midpoint is zero.	B1	
			Total	**8**	

Question			Answer/Indicative content	Marks	Guidance
11			B	1	
			Total	1	